AWS RDS PostgreSQL Database

Endpoint: quickbite.cny4u4aekmco.us-east-2.rds.amazonaws.com

Port: 5432

DB name: quickbite

Username: mansi1611 or quickbite.cny4u4aekmco.us-east-2.rds.amazonaws.com

Password: manshiv2516

S3 Bucket link: https://us-east-2.console.aws.amazon.com/s3/buckets/my-quickbite-bucket?region=us-east-2&bucketType=general

SQL DDL (Data Definition Language) Script for creating tables:

```sql
-- Create customers table
CREATE TABLE customers (
    customer_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    email VARCHAR(100) UNIQUE NOT NULL,
    phone VARCHAR(20)
);

-- Create menuitems table
CREATE TABLE menuitems (
    menuitem_id SERIAL PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    price DECIMAL(10, 2) NOT NULL,
    category VARCHAR(50)
);

-- Create orders table
CREATE TABLE orders (
    order_id SERIAL PRIMARY KEY,
    customer_id INT NOT NULL,
    order_date TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    total_amount DECIMAL(10, 2),
    FOREIGN KEY (customer_id) REFERENCES customers(customer_id)
);

-- Create orderitems table
CREATE TABLE orderitems (
    orderitem_id SERIAL PRIMARY KEY,
    order_id INT NOT NULL,
    menuitem_id INT NOT NULL,
    quantity INT NOT NULL CHECK (quantity > 0),
    FOREIGN KEY (order_id) REFERENCES orders(order_id),
    FOREIGN KEY (menuitem_id) REFERENCES menuitems(menuitem_id)
);
```

Orders table data points

```sql
-- insert sample orders
INSERT INTO orders (order_id, customer_id, order_date, total_amount) VALUES
(1, 1, '2025-05-01', 15.97),
(2, 2, '2025-05-02', 13.48),
(3, 3, '2025-05-03', 7.98),
(4, 5, '2025-05-04', 15.97),
(5, 20, '2025-05-06', 13.48),
(6, 30, '2025-05-06', 7.98),
(7, 10, '2025-05-07', 15.97),
(8, 24, '2025-05-02', 13.48),
(8, 23, '2025-05-04', 7.98),
(10, 17, '2025-05-09', 15.97),
(11, 23, '2025-05-10', 13.48),
(12, 12, '2025-05-02', 7.98);
```

Basic Queries:

1. **Get All Menu Items**:

```sql
SELECT
  name,
  price,
  category
FROM
  menuitems;
```

SELECT name, price, categor | Enter a SQL expressi

	name	price	category
1	Cheeseburger	8.99	Main Course
2	Veggie Pizza	11.49	Main Course
3	Samosa	6.75	Appetizer
4	Cheese Sticks	7.99	Appetizer
5	Chocolate Cake	4.5	Dessert
6	Iced Coffee	3.25	Beverage
7	Lemonade	2.99	Beverage

2. **Find Orders by Date:**

```sql
SELECT
    orders.order_id,
    customers.name AS customer_name,
    orders.total_amount
FROM
    orders
JOIN
    customers ON orders.customer_id = customers.customer_id
WHERE
    orders.order_date = '2025-05-08';
```

orders(+) 1

SELECT orders.order_id, cust | Enter a SQL expressio

	order_id	customer_name	total_amount
1	8	James Martinez	15.98
2	9	Sophia Hernandez	6.24

Intermediate Queries:

1. Write an SQL query to display the **CustomerName**, **OrderID**, **MenuItemName**, and **Quantity** for each item in the orders placed by **one customer**

```sql
select
	c.name as customer_name,
	o.order_id,
	m.name as menu_item_name,
	oi.quantity
from orders o
join customers c on o.customer_id = c.customer_id
join orderitems oi on o.order_id = oi.order_id
join menuitems m on oi.menuitem_id = m.menuitem_id
where c.customer_id = 2;
```

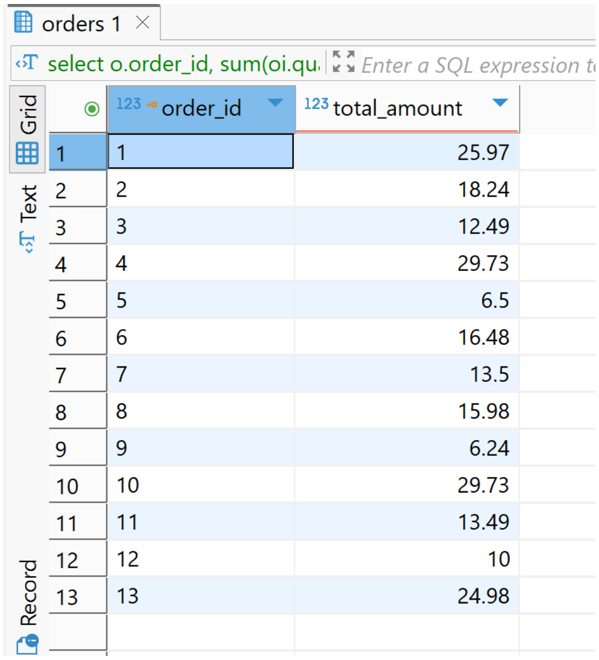
customers(+) 1

select c.name as customer_r | Enter a SQL expression to filter results (use Ctrl+Spa

	customer_name	order_id	menu_item_name	quantity
1	Jane Smith	2	Veggie Pizza	1
2	Jane Smith	2	Samosa	1

2. Calculate Total Order Value:Write an SQL query to calculate the total value of each Order. This should be based on the MenuItems ordered (i.e., Quantity * Price) and should show the OrderID and TotalAmount.

```sql
select
	o.order_id,
	sum(oi.quantity * m.price) as total_amount
from orders o
join orderitems oi on o.order_id = oi.order_id
join menuitems m on oi.menuitem_id = m.menuitem_id
group by o.order_id
order by o.order_id;
```



	order_id	total_amount
1	1	25.97
2	2	18.24
3	3	12.49
4	4	29.73
5	5	6.5
6	6	16.48
7	7	13.5
8	8	15.98
9	9	6.24
10	10	29.73
11	11	13.49
12	12	10
13	13	24.98

3. **Customer's Total Spend:** Write an SQL query to calculate the total amount spent by each customer, based on their Orders.

```
select
	c.customer_id,
	c.name as customer_name,
	sum(oi.quantity * m.price) as total_spent
from customers c
join orders o on c.customer_id = o.customer_id
join orderitems oi on o.order_id = oi.order_id
join menuitems m on m.menuitem_id = oi.menuitem_id
group by c.customer_id, c.name
order by total_spent desc;
```

customers 1

select c.customer_id, c.name

	customer_id	customer_name	total_spent
1	4	Bob Williams	29.73
2	10	William Lopez	29.73
3	1	John Doe	25.97
4	13	Mia Thomas	24.98
5	2	Jane Smith	18.24
6	6	Michael Davis	16.48
7	8	James Martinez	15.98
8	7	Olivia Garcia	13.5
9	11	Isabella Wilson	13.49
10	3	Alice Johnson	12.49
11	12	Alexander Anderson	10
12	5	Emma Brown	6.5
13	9	Sophia Hernandez	6.24

Advanced Tasks:

1. **Popular Menu Items**:Write an SQL query to find the most ordered **MenuItem** based on the **quantity ordered**. Display the **MenuItemName** and the **total quantity** ordered for each item.

```
select
	m.name as menu_item_name,
	sum(oi.quantity) as total_quantity_ordered
from orderitems oi
join menuitems m on oi.menuitem_id = m.menuitem_id
group by m.name
order by total_quantity_ordered desc;
```

menuitems 1

select m.name as menu_iter

	menu_item_name	total_quantity_ordered
1	Samosa	6
2	Veggie Pizza	6
3	Cheeseburger	5
4	Chocolate Cake	4
5	Iced Coffee	4
6	Cheese Sticks	4
7	Lemonade	2

2. **Revenue Breakdown by Customer**: Write an SQL query to calculate **total revenue** per **Customer**. Show the **CustomerName**, **Total Revenue**, and **Total Number of Orders**.

```sql
select
	c.name as customer_name,
	count(distinct o.order_id) as total_orders,
	sum (oi.quantity * m.price) as total_revenue
from customers c
join orders o on c.customer_id = o.customer_id
join orderitems oi on o.order_id = oi.order_id
join menuitems m on oi.menuitem_id = m.menuitem_id
group by c.name
order by total_revenue desc;
```

customers 1

select c.name as customer_r | Enter a SQL expression to filter results (use Ctrl+Spa

	customer_name	total_orders	total_revenue
1	William Lopez	1	29.73
2	Bob Williams	1	29.73
3	John Doe	1	25.97
4	Mia Thomas	1	24.98
5	Jane Smith	1	18.24
6	Michael Davis	1	16.48
7	James Martinez	1	15.98
8	Olivia Garcia	1	13.5
9	Isabella Wilson	1	13.49
10	Alice Johnson	1	12.49
11	Alexander Anderson	1	10
12	Emma Brown	1	6.5
13	Sophia Hernandez	1	6.24

3. **Update Price of Menu Item**: Write an SQL query to **update the price** of the **Cheeseburger** (MenuItemID = 1) by **10%**.

```sql
update menuitems
set price = price * 1.10
where name = 'Cheeseburger';
```

4. **Delete Specific Orders**: Write an SQL query to **delete all orders** placed by **Jane Smith** (CustomerID =2).

```sql
delete from orderitems   -- Deleting from orderitems table
where order_id in(
	select order_id from orders where customer_id = 3
);
```

```sql
delete from orders  -- Deleting from orders table
where customer_id = 3;
```

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